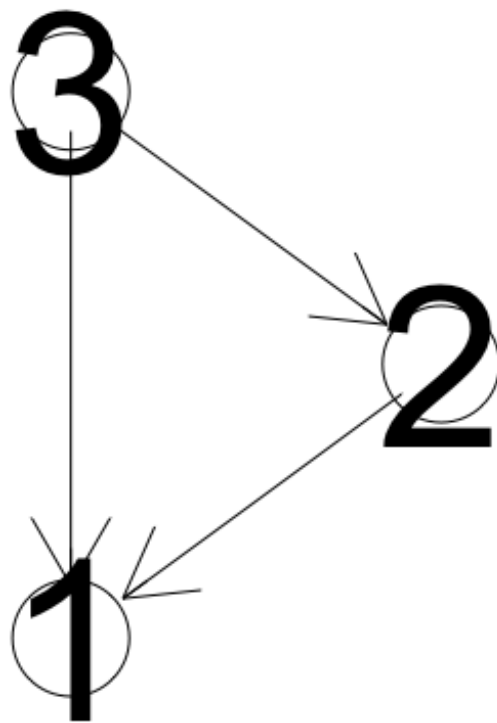
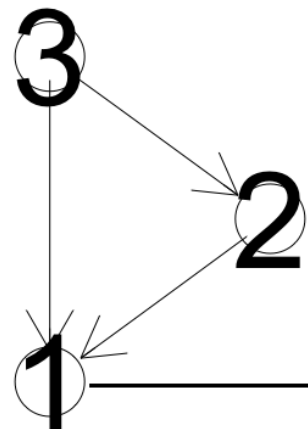


Example 1

True DAG



True DAG

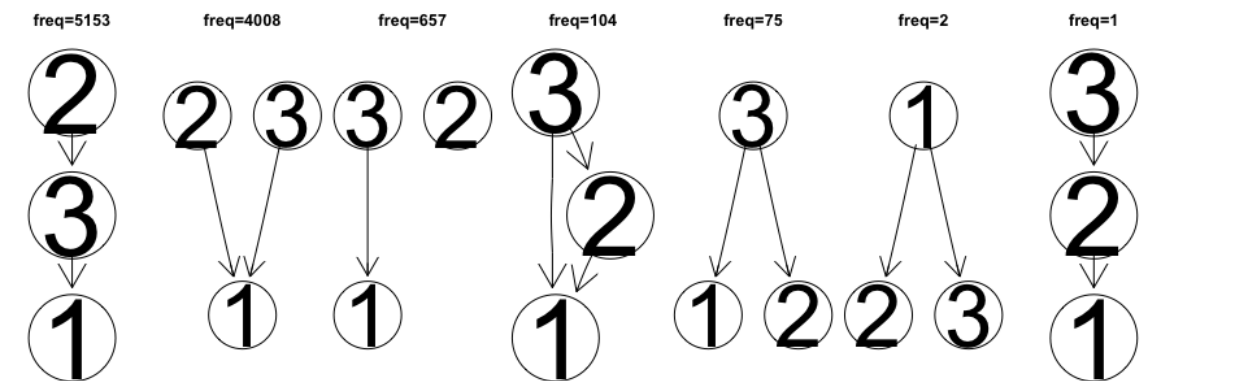
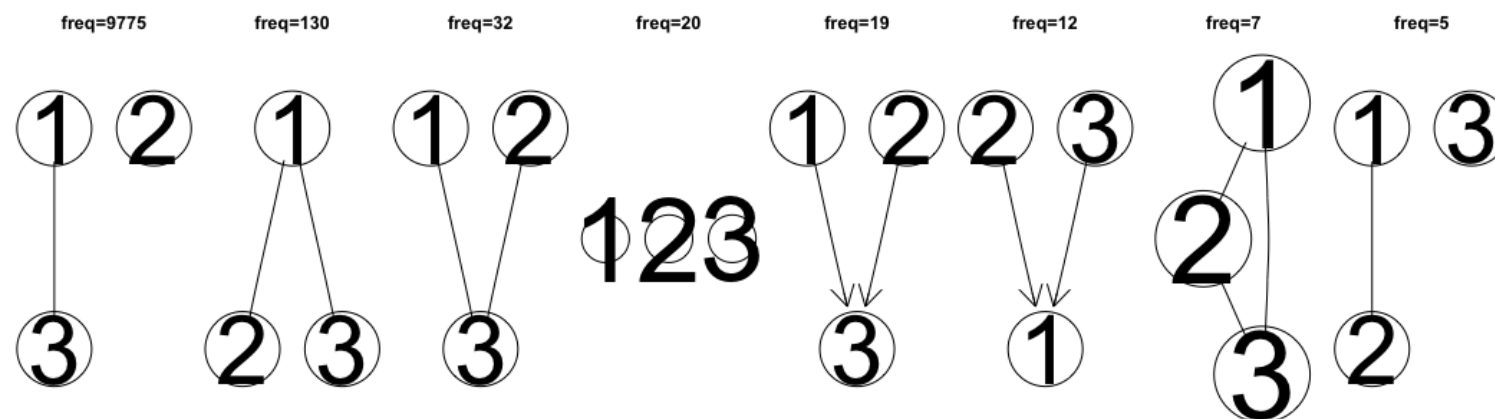


- Data size n : 20
- Number of iterations: 10000

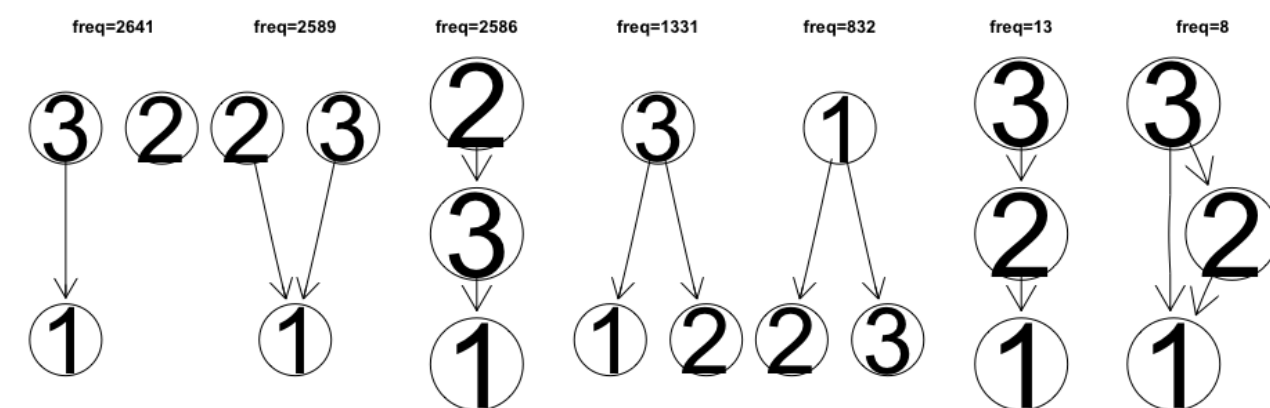
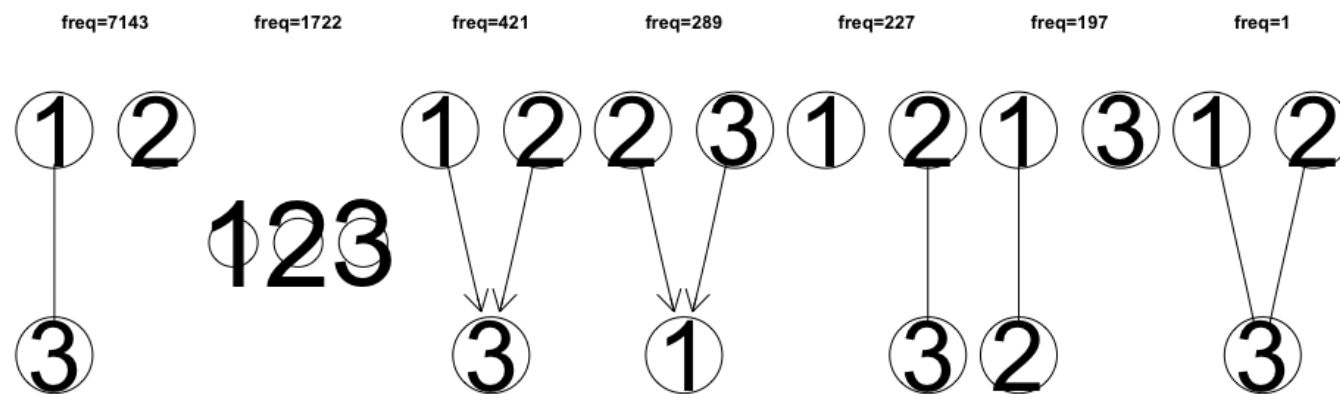
PC (alpha = 0.01)

BOSS (discount = 1)

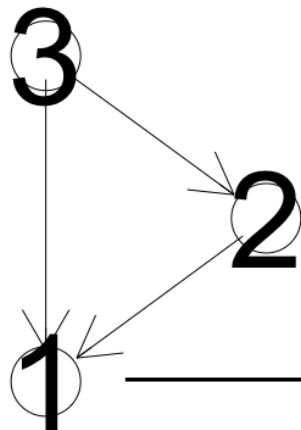
BOOTSTRAP



MCMC (95% CI)



True DAG



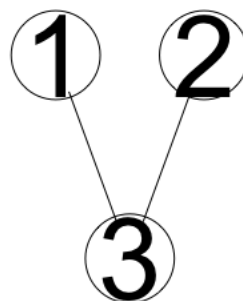
- Data size n : 200
- Number of iterations: 10000

PC (alpha = 0.01)

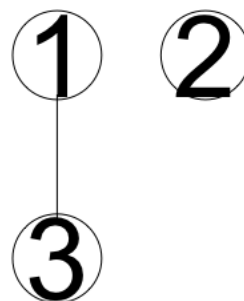
BOSS (discount = 1)

BOOTSTRAP

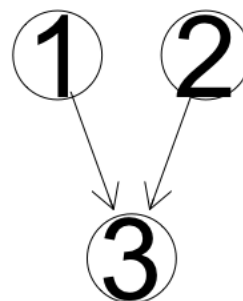
freq=4539



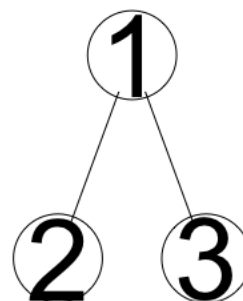
freq=4291



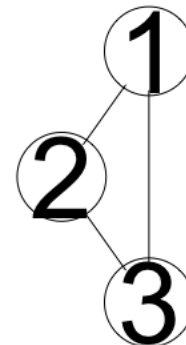
freq=1107



freq=51



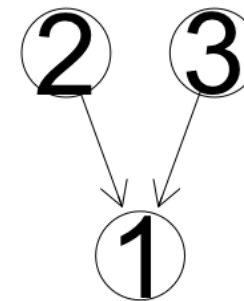
freq=12



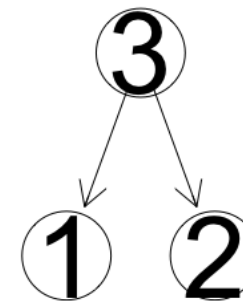
freq=8886



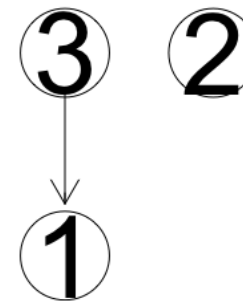
freq=607



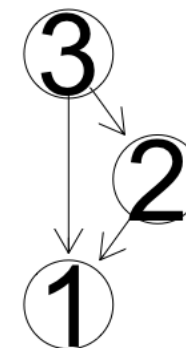
freq=354



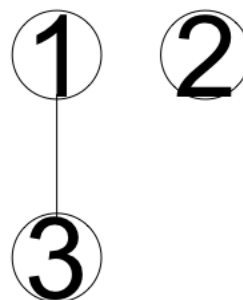
freq=82



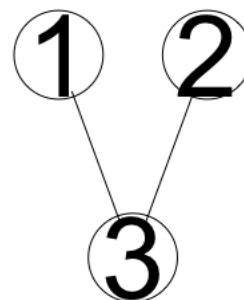
freq=71



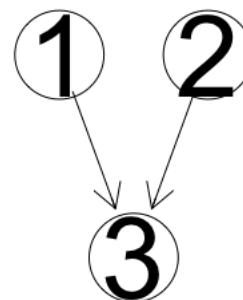
freq=4960



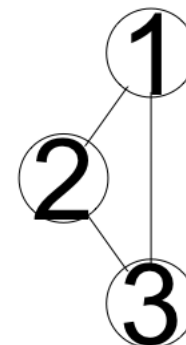
freq=3289



freq=1741



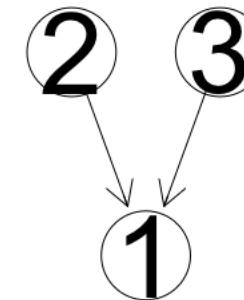
freq=10



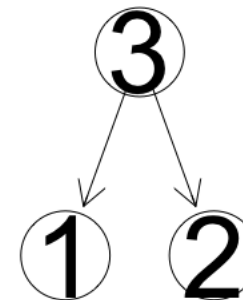
freq=7376



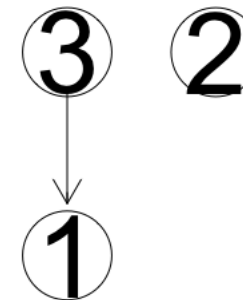
freq=1452



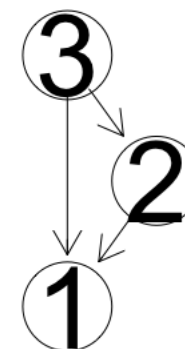
freq=755



freq=336

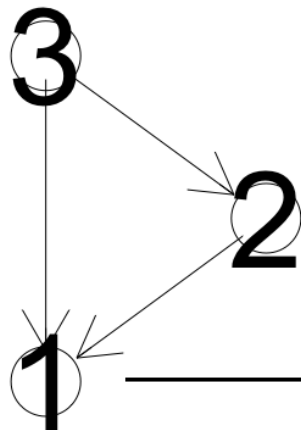


freq=81



MCMC (95% CI)

True DAG

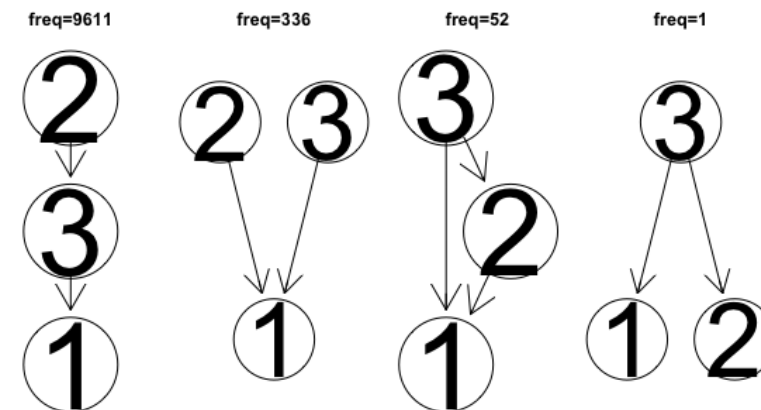
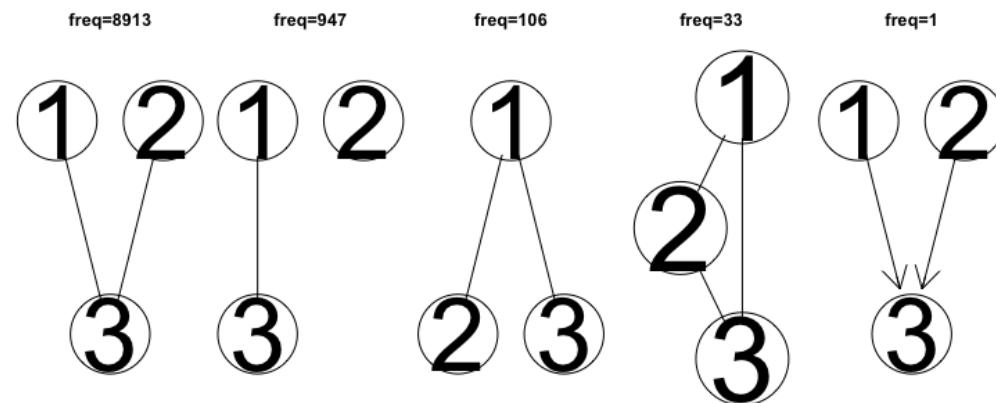


- Data size n : 500
- Number of iterations: 10000

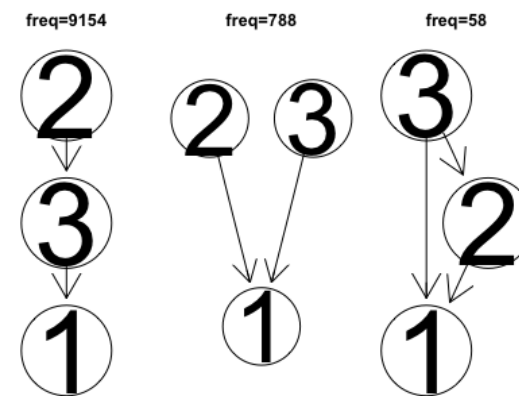
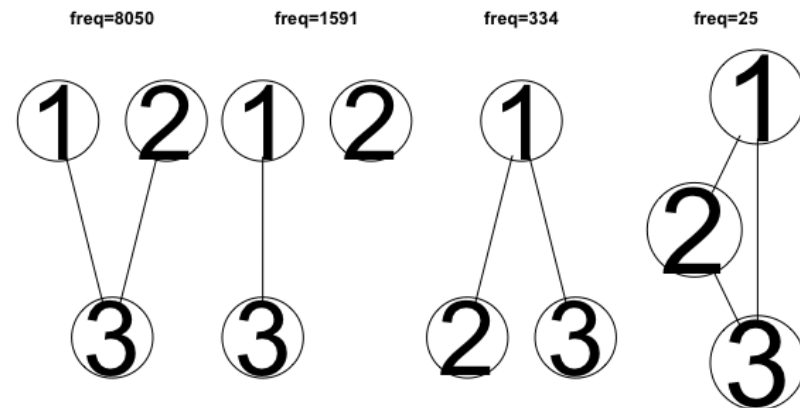
PC (alpha = 0.01)

BOSS (discount = 1)

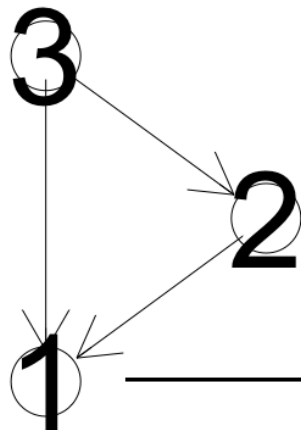
BOOTSTRAP



MCMC (95% CI)



True DAG



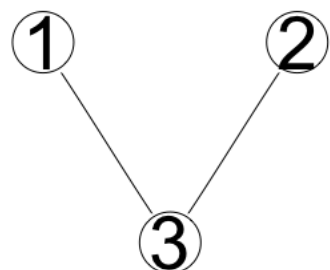
- Data size n : 2000
- Number of iterations: 10000

PC (alpha = 0.01)

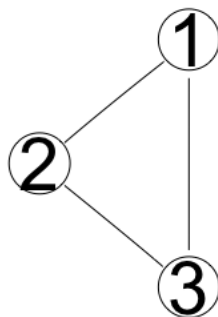
BOSS (discount = 1)

BOOTSTRAP

freq=9895



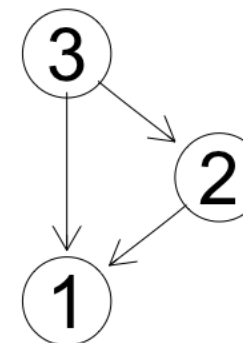
freq=105



freq=9942

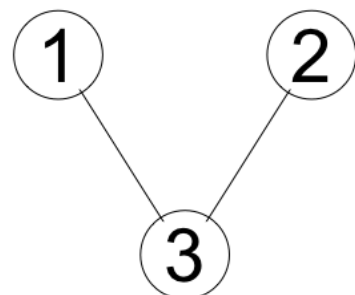


freq=58

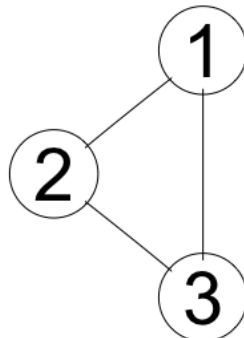


MCMC (95% CI)

freq=9924



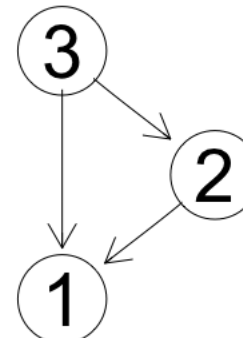
freq=76



freq=9993

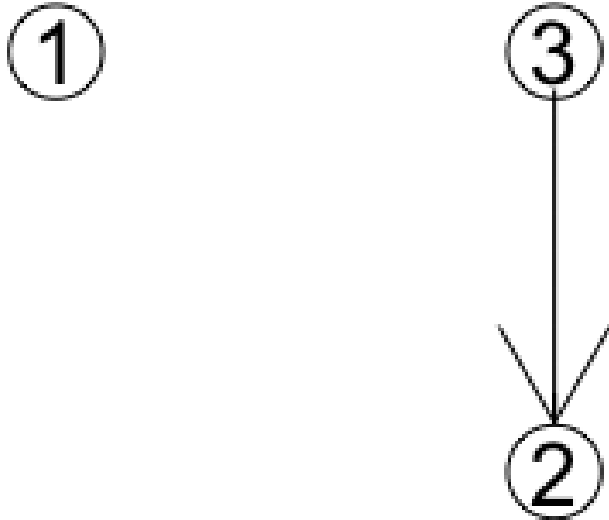


freq=7



Example 2

True DAG

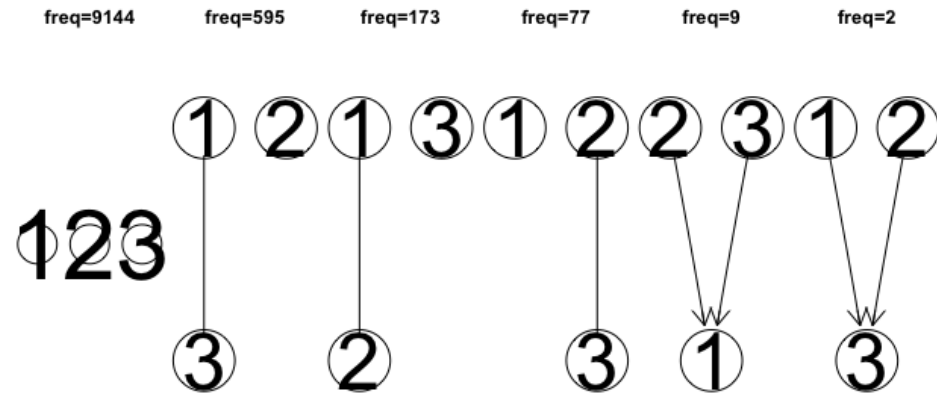


- Data size n : 20
- Number of iterations: 10000

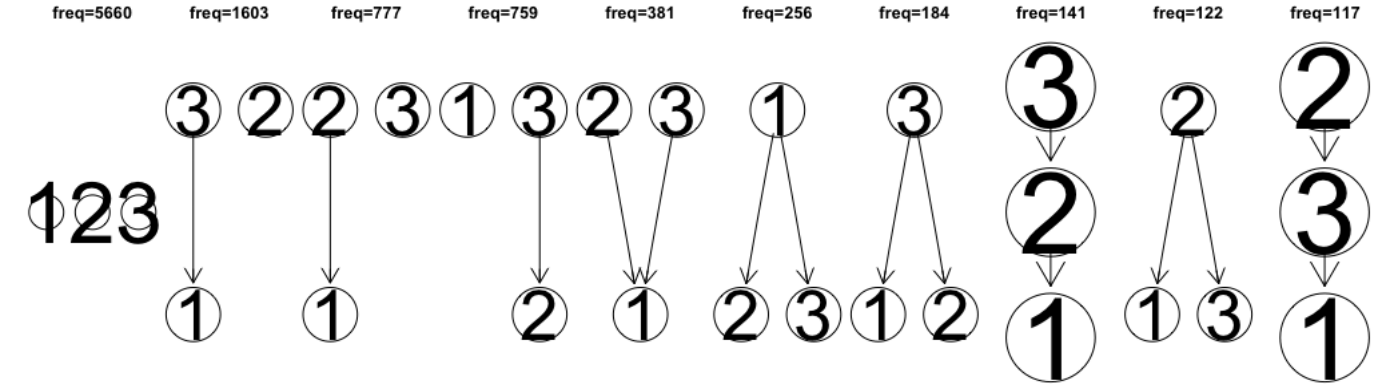
PC ($\alpha = 0.01$)

BOSS (discount = 1)

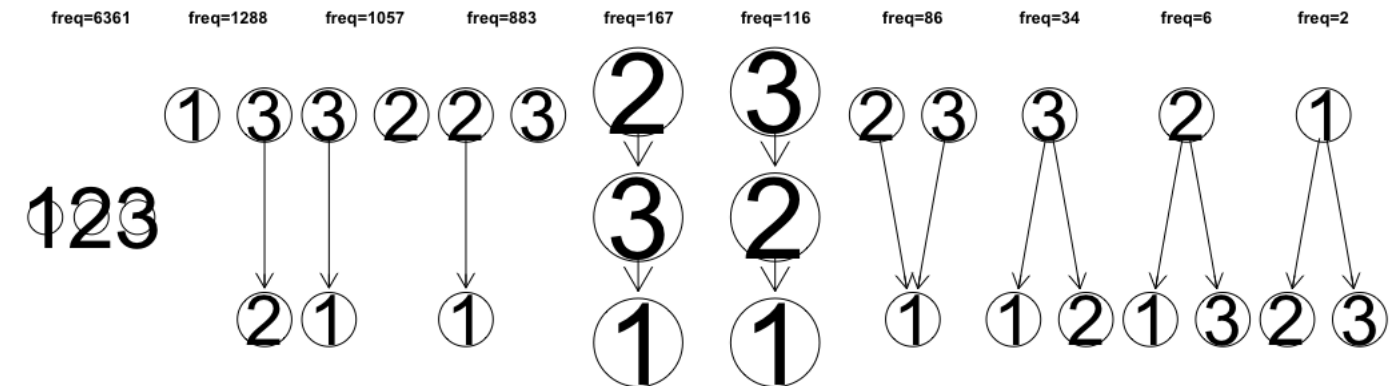
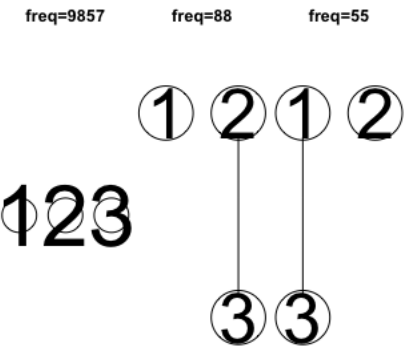
BOOTSTRAP



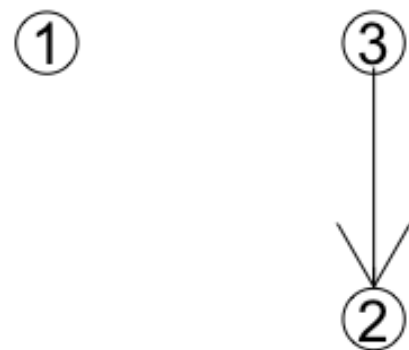
123



MCMC (95% CI)



True DAG

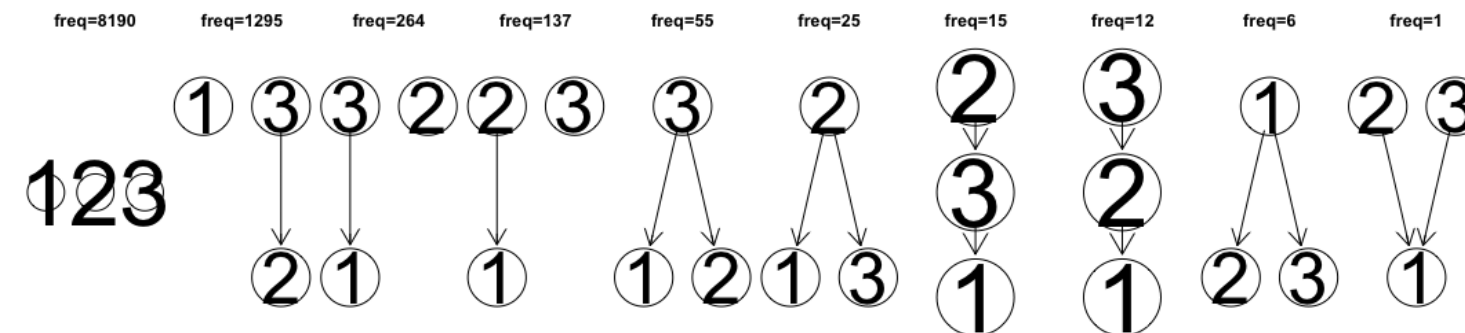
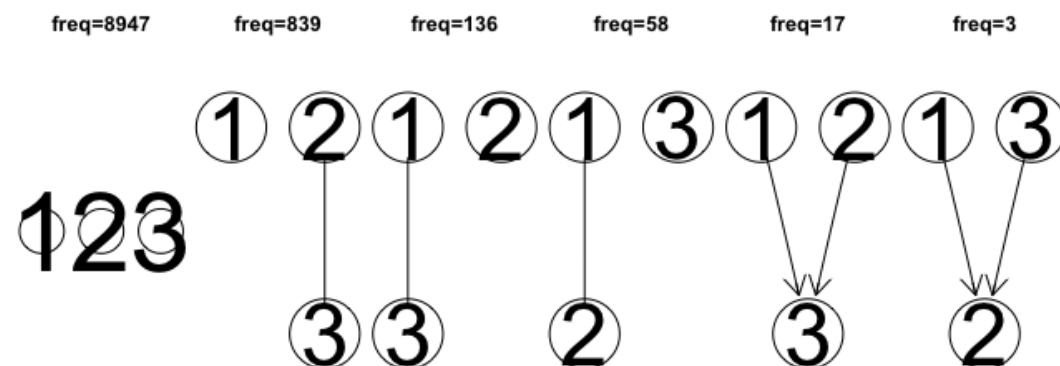


- Data size n : 200
- Number of iterations: 10000

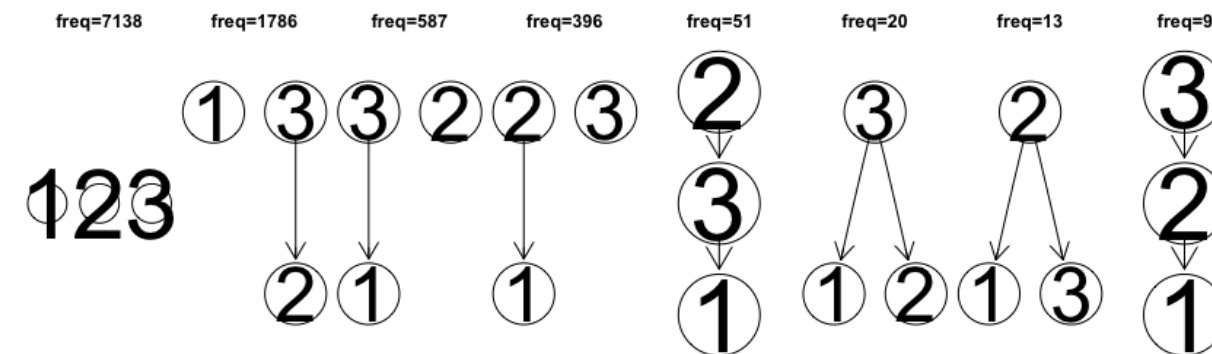
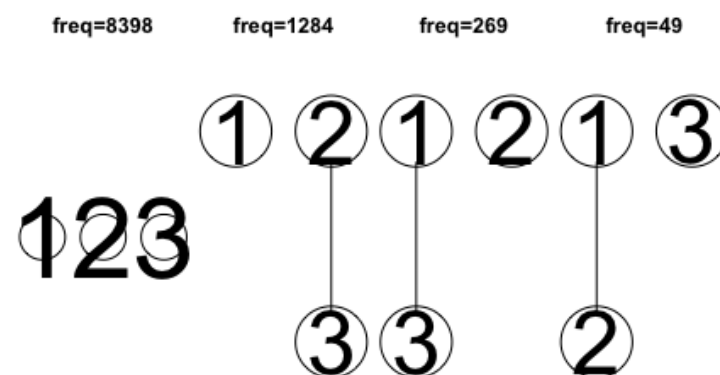
PC (alpha = 0.01)

BOSS (discount = 1)

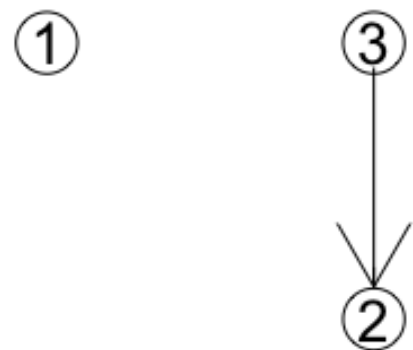
BOOTSTRAP



MCMC (95% CI)



True DAG

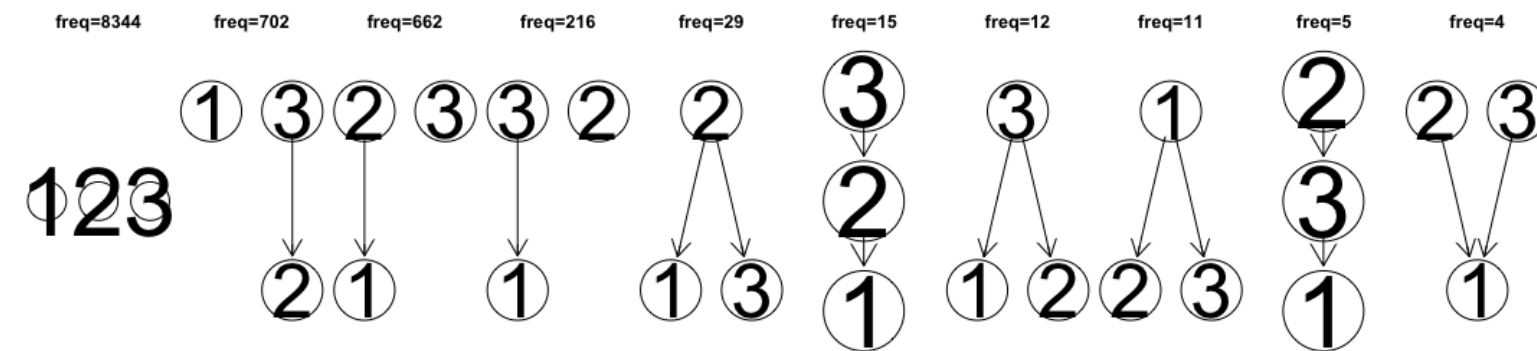
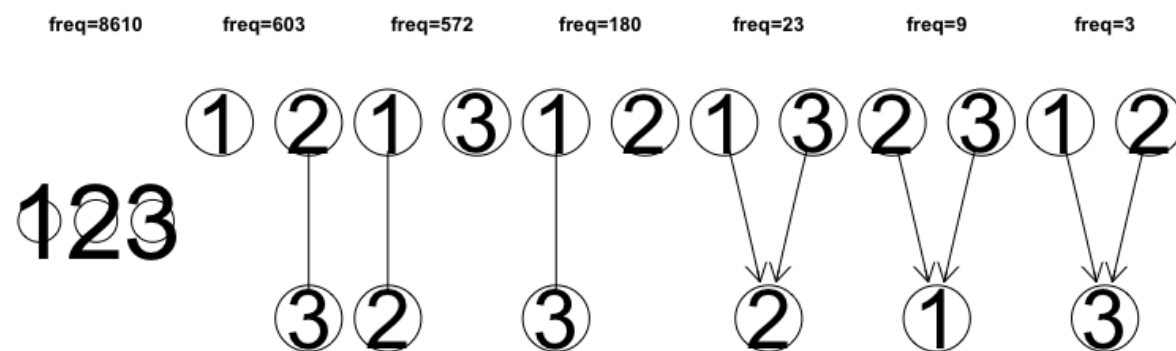


- Data size n : 500
- Number of iterations: 10000

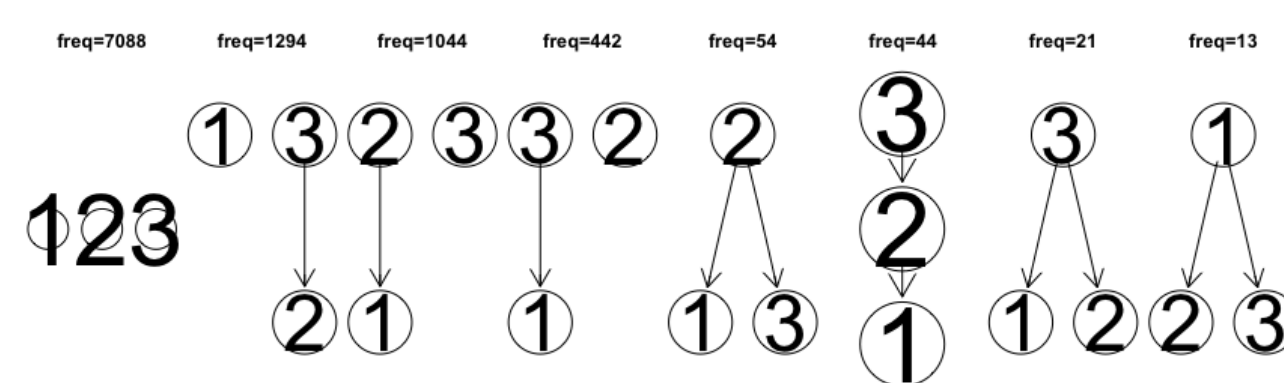
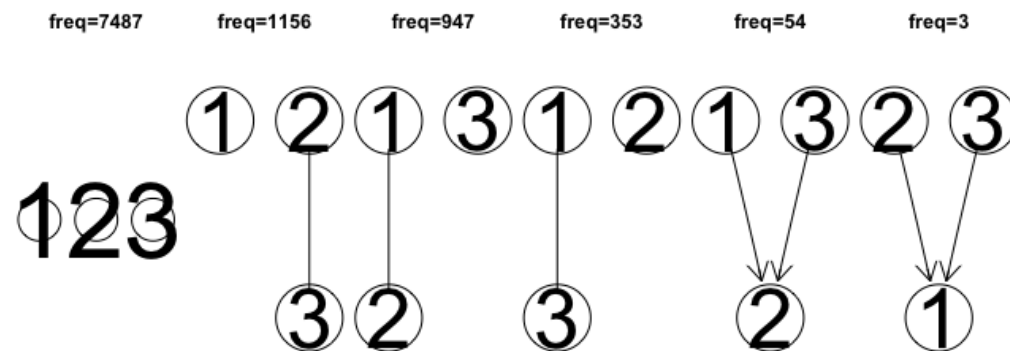
PC (alpha = 0.01)

BOSS (discount = 1)

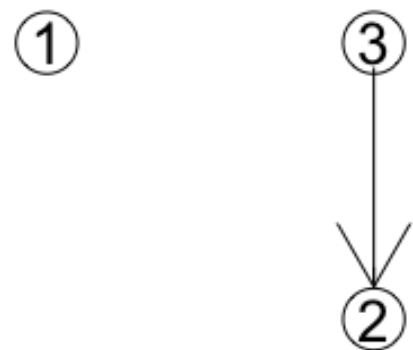
BOOTSTRAP



MCMC (95% CI)



True DAG

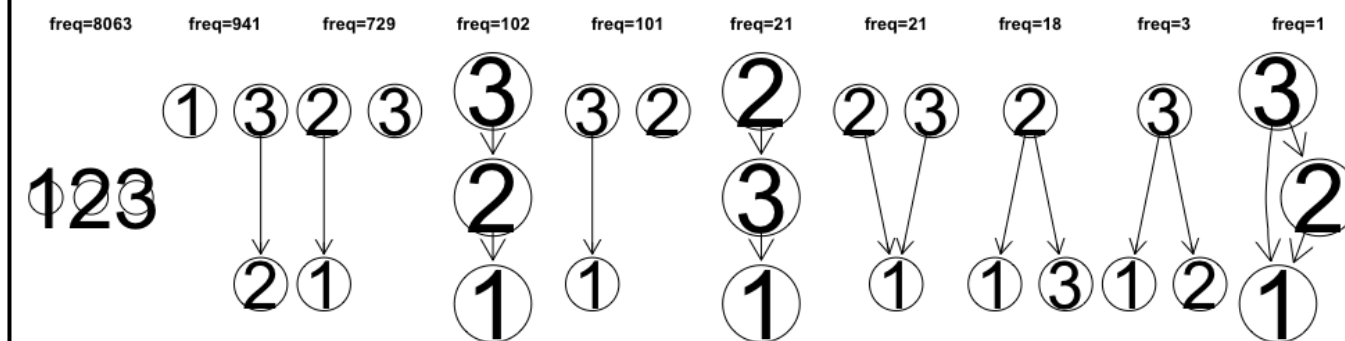
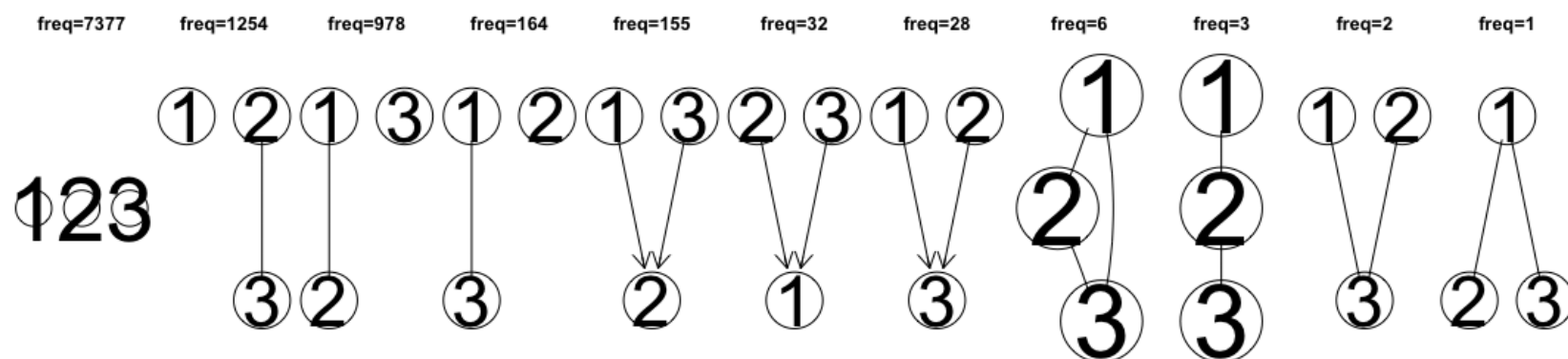


- Data size n : 2000
- Number of iterations: 10000

PC (alpha = 0.01)

BOSS (discount = 1)

BOOTSTRAP



MCMC (95% CI)

